

Code-Layout, Readability and Reusability

Date	15 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the code quality of the Human Development Index (HDI) Prediction System, focusing on code structure, readability, and reusability for easy maintenance and scalability.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation maintained throughout the code.
2	Proper File Structure	Files and folders are logically organized	Yes	Files organized into templates, static, model, and source files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear and descriptive variable names used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions are named based on their purpose.
5	Code Comments	Inline and block comments explain log	Partial	Comments added for important sections only.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code divided into reusable modules/functions.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate code removed wherever possible.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation and exception handling implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Data Processing	Python ,Pandas	Model Training	High
2	Prediction Module	Python , Scikit-learn	Prediction system	High
3	Web Interface	HTML , CSS	User Interface	Medium
4	Flask Backend	Python , Flask	Request Handling	High
5	Visualization	Matplotlib	Result Analysis	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well – structured project.
Readability	5	Easy to understand.
Reusability	4	Modules can be reused.
Documentation / Comments	4	Sufficient comments provided.
Overall Score	4.5 / 5	Good code quality and maintainability.